

Functional Specification: PLPI Batch Record Automation Phase 2
PLPI/BAR/FS/01/v1

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Revision History

Version	Previous version	Reason for revision	Issued
1	NA	Functional Specification expanded to cover all visible wireframe fields, buttons, tables, common printing controls, validations, document access, audit behaviour and route-specific processing.	Aug 2026

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1. Introduction

As defined in the approved URS, PLPI Batch Record Automation Phase 2 will digitalise the workflow from B&S Batch Add and electronic BAR creation through route-based printing and Leaflet Folding, while retaining controlled GMP checks, attribution and audit history.

1.1 Purpose of the document

The purpose of this document is to define the functional specification for PLPI Batch Record Automation Phase 2 and describe the functional behaviour required to meet the approved URS.

1.2 Scope

This document covers B&S Batch Add and BAR creation, BAR verification and digital line clearance, the Printer module, Label Printing, Leaflet Printing, Carton Issuing, Braille Printing where required, and Leaflet Folding.

Out of Scope: Pre-Assembly, Production Control, assembly-room activities, Post-Assembly QC, Pre-QP and QP Approval are excluded except for the controlled handoff from Leaflet Folding.

1.3 Abbreviations

These abbreviations are used throughout this document.

IT	Information Technology
URS	User Requirement Specification
DS	Design Specification
FS	Functional Specification
PLPI	Parallel Import / PLPI system workflow
BAR	Batch Assembly Record
PCL	Product Check Log
B&S	B&S Batch Add / batch identifier

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GMP	Good Manufacturing Practice
MFG	Manufacturing / manufacturer
QP	Qualified Person
RP / RPi	Responsible Person / Responsible Person import
QA	Quality Assurance
PDF	Portable Document Format
ECMA	European Centralised Marketing Authorisation

2. Overall Description

This section provides a general description of the Phase 2 workflow. An eligible checked batch is selected in B&S Batch Add, one electronic BAR is generated and signed after verification and digital line clearance, and the batch enters Label Printing. Where required, it continues through Leaflet Printing and then through Carton Issuing for reboxing or Braille Printing for applicable relabelling. When all route-specific printing is complete, the batch enters Leaflet Folding and is handed to the next configured process stage.

3. Functional Specification

This section describes the complete functional behaviour shown in the approved wireframe. Within the module tables, Existing identifies a field, table, screen or button replicated from the current PLPI system; Existing - updated function identifies a replicated control with new validation, electronic record, audit or routing behaviour; and New identifies functionality introduced by this change. Controls identified as common across all Printing Modules are specified once in section 3.2 and apply to Label Printing, Leaflet Printing and Braille Printing, with Carton Issuing using the applicable common queue, product, document and audit controls.

3.1 B&S Batch Add and BAR Creation Module

3.1	B&S Batch Add and BAR Creation Module
Priority	High

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3.1	B&S Batch Add and BAR Creation Module		
Purpose	To select eligible B&S batch records, create one controlled electronic BAR, complete BAR verification and digital line clearance, and release the signed batch to Label Printing.		
Role	Authorised B&S Batch Add / BAR Creation User		
Input	Field / Button / Table Column	Field type	Functional description
	Country	Existing field	Search filter replicated from the current B&S Batch Add screen. Filters eligible records by country.
	Site	Existing field	Search filter replicated from the current screen. Filters eligible records by processing site.
	Category	Existing field	Search filter replicated from the current screen. Values include Reboxing and Relabelling.
	Stock	Existing field	Existing stock filter retained on the search panel.
	Status	Existing field - updated values	Filters the grid by All, Not Printed BAR or Printed BAR status.
	Batch_No	Existing field	Existing batch-number search field used to locate a B&S batch.
	Search	Existing button	Applies the entered search criteria and refreshes the B&S Batch Add table.
	B&S Batch Add table	Existing table - updated function	Displays eligible source records and supports controlled row selection for BAR generation.

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3.1	B&S Batch Add and BAR Creation Module		
	Select	Existing table control - updated function	Checkbox used to select one or more compatible source rows. Incompatible product, batch or expiry combinations shall be rejected.
	Product Status	Existing table field	Displays the current product or batch status.
	Site	Existing table field	Displays the processing site.
	Country	Existing table field	Displays the product country.
	Part No.	Existing table field	Displays the existing product/component part number.
	Foreign Name	Existing table field	Displays the foreign product name.
	ECMA	Existing table field	Displays the approved ECMA/reference value.
	Strength	Existing table field	Displays the product strength.
	Pack Size	Existing table field	Displays the approved pack size.
	B&S Batch Number	Existing table field	Displays the controlled B&S batch identifier.
	Expiry Date	Existing table field	Displays the batch expiry date.
	Quantity	Existing table field	Displays the available/required batch quantity.
	IMP	Existing table field	Displays the current IMP indicator/value.
	Invoice	Existing table field	Displays the source invoice reference.
	Description	Existing table field	Displays the existing product description.

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3.1	B&S Batch Add and BAR Creation Module		
	Warehouse	Existing table field	Displays the source warehouse/location.
	Supplier	Existing table field	Displays the supplier associated with the record.
	PL No.	Existing table field	Displays the product licence number.
	Product ID	Existing table field	Displays the existing product identifier.
	MFG ID	Existing table field	Displays the manufacturing identifier.
	Action Count	Existing table field	Displays the current action count held against the record.
	OBJID	Existing table field	Displays the existing system object identifier.
	SPL Flag	Existing table field	Displays the existing SPL flag.
	Category	Existing table field	Displays the Reboxing or Relabelling route.
	Stock	Existing table field	Displays the existing stock value.
	Regulatory	Existing table field	Displays the existing regulatory status/value.
	Workflow Implemented	Existing table field	Displays whether the applicable workflow has been implemented.
	Legal Category	Existing table field	Displays the existing legal category.
	Generate BAR	Existing button - updated function	Replicated current-system action enhanced to create one populated controlled electronic BAR from the

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3.1	B&S Batch Add and BAR Creation Module		
			confirmed eligible selection.
	Print Batch Details	Existing button	Opens the selected batch-detail output for controlled printing.
	Total BNS Batch	Existing indicator	Displays the total number of B&S batches in the current result/context.
	Total Composite Batch	Existing indicator	Displays the total number of composite batches.
	Total Packs	Existing indicator	Displays the total pack quantity.
	Not Printed BAR	Existing status button/indicator	Identifies records for which the BAR has not been printed/generated.
	Generated BAR	New controlled document	Provides a populated 13-page electronic BAR with batch identity, module records, evidence areas and audit history.
	View Generated BAR	New controlled button	Opens the generated BAR in view-only mode after generation.
	Print Generated BAR	New controlled button	Prints the populated controlled BAR and records the user and date/time in the audit history.
	PCL completeness/invoice check	New electronic check - existing process	Confirms the PCL is complete and the invoice is attached.
	Batch/expiry check	New electronic check - existing process	Confirms the batch number and expiry date are correct.

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3.1	B&S Batch Add and BAR Creation Module		
	Product/pack/reference check	New electronic check - existing process	Confirms product name, strength, pack size and ECMA are correct.
	Comments	New field	Captures relevant line-clearance or BAR comments.
	BAR Created By / Created Date-Time	New audit fields	Displays the authenticated BAR creator and creation timestamp.
	User Sign Off	New controlled button	Captures the authenticated user/date-time, locks the B&S checks and moves the batch to Label Printing.
Operations	The user opens the existing B&S Batch Add screen, enters any required Country, Site, Category, Stock, Status and Batch_No criteria and selects Search. The existing results table displays the complete source-record data. The user selects one eligible row or compatible source rows and selects Generate BAR. The system validates eligibility and compatibility, creates one populated controlled BAR and retains the link to every selected source record. The BAR contains 13 controlled pages covering the BAR cover, B&S line clearance, Label Printing, Leaflet Printing, label evidence, Braille Printing, Leaflet Folding, Carton Issuing, downstream stage placeholders and the BAR audit trail. For reboxing, the completed Change of Pack Size form is added as an unnumbered attachment. The user may select Print Batch Details, View Generated BAR or Print Generated BAR as applicable. The user completes the three electronic B&S checks, adds comments where required and selects User Sign Off. Sign-off records the user/date-time, locks the checks, removes the batch from the active B&S Batch Add queue and makes it available in Label Printing.		
Use case	A BAR Creation user finds an eligible checked batch, confirms the batch and route data, generates the BAR, completes line clearance and signs the record for printing.		
Output	Filtered B&S Batch Add result; selected-source traceability; populated 13-page electronic BAR and applicable reboxing		

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3.1	B&S Batch Add and BAR Creation Module
	attachment; printed batch details where requested; completed line-clearance checks and comments; BAR creation/printing/sign-off audit events; and a signed batch in the Label Printing queue.
DS ID	3.1
URS ID	4.1.2 - 4.1.7
Business Rule	Search and table fields replicated from the current system remain Existing. BAR population, electronic checks, audit attribution and automated queue handoff are New or updated functions. Generate BAR requires at least one eligible selection. Multiple selections must have compatible product, batch and expiry data. A batch shall have only one active controlled BAR. View/Print Generated BAR shall be unavailable before BAR generation. User Sign Off shall be unavailable until all mandatory line-clearance checks are complete. Completed records shall not be regenerated through the normal process.

3.2 Printing Module - Common Controls

3.2	Printing Module - Common Controls		
Priority	High		
Purpose	To provide a single Printer entry point, route-controlled work queues and common document, quantity, completion and audit controls for Label Printing, Leaflet Printing, Carton Issuing and Braille Printing.		
Role	Authorised Printer User		
Input	Field / Button / Table Column	Field type	Functional description
	Printer Menu	Existing screen/menu - updated function	Common entry screen containing Label Printing, Leaflet Printing, Carton Issuing and Braille Printing options.
	Label Printing	Existing module button/tile	Opens the Label Printing queue.
	Leaflet Printing	Existing module button/tile	Opens the Leaflet Printing queue.
	Carton Issuing	Existing module button/tile	Opens the Carton Issuing queue.

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3.2	Printing Module - Common Controls		
	Braille Printing	Existing module button/tile	Opens the Braille Printing queue.
	Active / In Progress counts	New calculated indicators	Displays the current Active and In Progress workload for each Printer option.
	B&S Batch Number	Existing search field	Common queue search field used in all printing modules.
	MFG Lot No.	Existing search field	Common queue search field used in all printing modules.
	Search	Existing button	Applies the common queue search criteria.
	Printing queue table	Existing table - updated function	Common table layout used by Label, Leaflet, Carton and Braille queues; eligibility and status are controlled by module.
	B&S Batch Number	Existing queue column	Displays the controlled B&S batch identifier.
	MFG Lot No.	Existing queue column	Displays the manufacturing lot number.
	Product Name	Existing queue column	Displays the product name.
	Strength	Existing queue column	Displays product strength.
	Pack Size	Existing queue column	Displays pack size.
	ECMA	Existing queue column	Displays the approved reference.
	Expiry Date	Existing queue column	Displays batch expiry.
	Required Qty	Existing data - module-specific value	Displays label quantity, leaflet quantity, carton quantity or braille quantity according to the selected queue.
	Status	New calculated queue column	Displays Active, In Progress or the applicable downstream status.
	Select queue row	Existing table action -	Opens the selected batch in the applicable printing detail screen.

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3.2	Printing Module - Common Controls		
		updated function	
	Back to Batch Queue	Existing button	Returns from a printing detail screen to its batch queue without completing the stage.
	Product Information panel	Existing display - common	Common read-only panel shown in Label, Leaflet, Carton, Braille and Leaflet Folding details.
	Product Name / Foreign Name	Existing fields - common	Displays the approved product names.
	Strength / Country of origin	Existing fields - common	Displays strength and source country.
	Pack Size / Units per pack	Existing fields - common	Displays packaging configuration.
	B&S Batch Number / MFG Lot No.	Existing fields - common	Displays batch identifiers.
	ECMA / PL No.	Existing fields - common	Displays approved ECMA and product licence references.
	Expiry Date / Quantity	Existing fields - common	Displays expiry and approved batch quantity.
	Product Introduced	Existing field - common	Displays the product introduction date/value.
	Leaflet Date / Date Revised	Existing fields - common	Displays the leaflet date and latest revision date.
	Batch Documents panel	New controlled access - common	Provides controlled document buttons for each printing module.
	View Carton Artwork	Existing document button - common	Opens approved carton artwork.
	View Peel Artwork	Existing document	Opens approved peel/label artwork.

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3.2	Printing Module - Common Controls		
		button - common	
	View Braille Artwork	Existing document button - common	Opens approved braille artwork.
	View Mock-up	Existing document button - common	Opens the approved mock-up.
	View BAR	Existing document button - updated access	Opens the applicable controlled BAR/reference document.
	Print Generated BAR	New controlled button - common	Opens the populated BAR and permits controlled printing with audit capture.
	Continuation Page panel	New controlled access - common	Provides supporting continuation documents associated with the batch.
	Print Label Attachment	New controlled button - common	Prints the label-evidence attachment page.
	Print BAR Continuation	New controlled button - common	Prints the BAR continuation page.
	Print Cold-chain Page	New controlled button - common	Prints the cold-chain record page.
	Print Change of Pack Size	New route-specific button	Available for reboxing and prints the completed Change of Pack Size attachment.
	Print	Existing button - common	Existing replicated module button. Opens the new common Print pop-up for the selected label, leaflet or braille line.
	Print pop-up	New pop-up - common	The complete pop-up opened from Print is new. It provides all new quantity, category, reason,

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3.2	Printing Module - Common Controls		
			preview and submission controls defined below.
	Quantity Needed	New read-only field - common	Displays the approved quantity for the selected component.
	Quantity to Print	New entry field - common	Requires a whole-number print quantity greater than zero.
	Category	New controlled selection - common	Values are Test Print, Actually Print and Extra Print.
	Reason	New conditional field - common	Enabled and mandatory for Extra Print. Values include Print damage, Line setup waste, Reconciliation correction, Printing alignment check and Supervisor approved extra.
	Print (dialog)	New controlled button - common	Sends the selected quantity/category to the applicable print workflow and records the event.
	Close print preview	New button - common	Closes the print preview without submitting a print action.
	Line Completion	New status/action - common	Shows Pending, Mark as Done or Done and records the completing user/date-time.
Operations	The authorised user opens the existing Printer menu and selects a printing option. All four printing modules use the same queue search, queue columns, product-information panel, controlled-document panel, Back to Batch Queue action, user attribution and audit principles. Label, Leaflet and Braille Printing also share the unified Print window. The user enters Quantity to Print, selects Test Print, Actually Print or Extra Print and selects Print. Extra Print enables Reason and cannot continue without a selected reason. The system records item, quantity needed, quantity printed, category, reason, user and date/time. Component completion remains separate from the print action. Print Done is enabled only when all mandatory lines for that module are Done. Carton Issuing uses the common		

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3.2	Printing Module - Common Controls
	batch/document/audit presentation but records carton issue rather than a printer run.
Use case	A Printer user selects the applicable printing option, opens an eligible batch, reviews the controlled documents, completes the required component lines and releases the batch to its next route-defined activity.
Output	Module workload counts; controlled printing queues; complete product and document context; print-run records by category; extra-print reasons; line-level status and attribution; module completion status; and route-controlled handoff.
DS ID	3.2
URS ID	4.2.1 - 4.2.3, 4.2.5, 4.2.8
Business Rule	Existing describes a replicated current-system screen, field or button. New describes additional electronic behaviour or data. Existing - updated function describes a replicated control whose validation, audit or routing is changed. Common controls shall behave consistently across Label, Leaflet and Braille Printing unless a module-specific rule states otherwise. Queue contents shall be route- and prior-stage-controlled. Extra Print requires a positive whole quantity and reason. All document openings, prints, line completions and Print Done actions shall be attributable and auditable. The entire Print pop-up and every field, selection and button within it are New functionality.

3.3 Label Printing Module

3.3	Label Printing Module		
Priority	High		
Purpose	To print and complete every label line required by the approved BAR route before the batch moves to Leaflet Printing or the next applicable stage.		
Role	Authorised Printer User		
Input	Field / Button / Table Column	Field type	Functional description
	Label Printing List	Existing queue - updated eligibility	Lists BAR-signed batches ready for Label Printing and uses all common Printer queue fields/buttons defined in section 3.2.

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3.3	Label Printing Module		
	Labels to Print table	Existing table - updated function	Displays every route-derived label component and its printing/completion state.
	Label	Existing table field	For Reboxing: Carton Label, End of Pack Label and Security Seals. For Relabelling: Blister / Pack Label and Obscure Label.
	Reference	Existing table field	Displays the approved ECMA/reference; Security Seals use the Seal reference.
	Size	Existing table field	Displays approved pack size or Standard for applicable labels.
	Quantity	Existing table field - calculated value	Displays the route-derived required quantity; Security Seals default to the controlled seal quantity.
	Location	Existing table field	Displays warehouse/component location.
	In Hand Quantity	Existing table field	Displays available stock/on-hand quantity.
	Print	Existing button - common updated function	Opens the common Print window described in section 3.2.
	Line Completion	New table status	Shows Pending or Done for the label line.
	Completed By / Date	New audit table field	Displays the authenticated completing user and date/time.
	Change of Pack Size form	Existing form - digitised/route-specific	Displayed for Reboxing. Captures Received As and Assembled As information and their sign-offs.
	Received As sign-off	New electronic control - existing process	Must be complete before label printing starts for Reboxing.
	Assembled As sign-off	New electronic control -	Must be complete before label printing starts for Reboxing.

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3.3		Label Printing Module	
		existing process	
	Print Done	New controlled button	Enabled after all displayed label rows are Done and finalises Label Printing.
	Back to Batch Queue	Existing button - common	Returns to the Label Printing queue.
Operations	The user opens an eligible BAR-signed batch from Label Printing. The common product and document panels are displayed. The system derives the table rows from the approved route. Reboxing displays Carton Label, End of Pack Label and Security Seals; Relabelling displays Blister / Pack Label and Obscure Label. For Reboxing, both Received As and Assembled As sections must be completed and electronically signed before any Print button is enabled. For each row the user verifies reference, size, quantity, location and stock, opens Print, completes the applicable Test Print/Actually Print/Extra Print action and completes the line. Done and the completing user/date-time are displayed. When all route-derived rows are Done, Print Done finalises the module and routes a leaflet-required batch to Leaflet Printing.		
Use case	A Printer user completes all route-derived label lines for an approved BAR and confirms the batch ready for the next printing activity.		
Output	Completed route-specific label table; Change of Pack Size record for reboxing; test, actual and extra-print history; line completion attribution; final Label Printing status; and handoff to Leaflet Printing or the next applicable stage.		
DS ID	3.3		
URS ID	4.2.4, 4.2.5, 4.2.8		
Business Rule	The common controls in section 3.2 apply. Label rows shall be route-derived. Reboxing printing is blocked until both Change of Pack Size sign-offs are complete. A test print shall be completed before the corresponding line may be completed where required. Required quantity shall be satisfied before Mark as Done/Done. Additional prints after completion shall be audited and shall not overwrite the original completion. Print Done requires all mandatory rows to be Done.		

3.4 Leaflet Printing Module

3.4	Leaflet Printing Module
Priority	High

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3.4	Leaflet Printing Module		
Purpose	To review the approved leaflet, complete the required test or master check, print each required leaflet line and route the batch according to the approved packaging process.		
Role	Authorised Printer User		
Input	Field / Button / Table Column	Field type	Functional description
	Leaflet Printing List	Existing queue - updated eligibility	Lists batches that completed Label Printing and require a leaflet; uses the common queue fields/buttons in section 3.2.
	Leaflets to Print table	Existing table - updated function	Displays required leaflet component lines and completion status.
	Leaflet	Existing table field	Identifies the leaflet component.
	Reference	Existing table field	Displays the approved ECMA/leaflet reference.
	Size	Existing table field	Displays the approved pack-size leaflet format.
	Quantity	Existing table field - calculated value	Displays the approved leaflet quantity.
	Location	Existing table field	Displays the leaflet stock location.
	Print	Existing button - common updated function	Opens the controlled leaflet PDF/print workflow and the common quantity/category controls.
	Mark as Done	New controlled button	Enabled only after an Actually Print event has been recorded for the leaflet line.
	Line Completion	New table status	Displays Mark as Done or Done.
	Completed By / Date	New audit table field	Displays the completing user and timestamp.

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3.4		Leaflet Printing Module	
	Print Done	New controlled button	Enabled only after every required leaflet line is Done.
	Back to Batch Queue	Existing button - common	Returns to the Leaflet Printing queue.
Operations	The user opens an eligible leaflet-required batch after Label Printing. The system displays the common product/document panels and the Leaflets to Print table. Selecting Print opens the approved controlled leaflet PDF and records the opening user/date-time. The user completes the required test/master review and selects the required print category and quantity. Mark as Done remains disabled until an Actually Print event exists for the line. Extra Print requires a reason. When the line is complete, the table records Done with user/date-time. Print Done finalises the module. Reboxing routes to Carton Issuing; Relabelling with braille required routes to Braille Printing; otherwise, the batch routes to Leaflet Folding or the next applicable stage.		
Use case	A Printer user reviews and prints the approved leaflet for the batch, completes every leaflet line and releases the batch to its route-specific preparation stage.		
Output	Approved leaflet/PDF access history; test/master evidence; actual and extra-print records; completed leaflet table; user/date-time attribution; final Leaflet Printing status; and route to Carton Issuing, Braille Printing or Leaflet Folding.		
DS ID	3.4		
URS ID	4.2.5, 4.2.6, 4.2.8		
Business Rule	The common controls in section 3.2 apply. Only leaflet-required batches that completed Label Printing shall appear. The approved PDF opening shall be audited. Test/master acceptance shall precede controlled completion where required. Mark as Done requires an actual-print record. Rejected master/test copies shall be retained as exception evidence. Print Done requires all leaflet rows to be Done. Routing shall use the approved Reboxing/Relabelling and braille-required values.		

3.5 Carton Issuing Module

3.5		Carton Issuing Module
Priority	High	

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3.5	Carton Issuing Module		
Purpose	To record the controlled issue of approved cartons for reboxing batches after Leaflet Printing.		
Role	Authorised Printer / Packaging User		
Input	Field / Button / Table Column	Field type	Functional description
	Carton Issuing List	Existing queue - updated eligibility	Lists Reboxing batches that completed Leaflet Printing and uses the common Printer queue fields/buttons in section 3.2.
	Carton to Issue table	Existing table - updated function	Displays the single controlled carton-issue line and completion status.
	Carton Reference	Existing table field	Displays ECMA/part number or the approved carton reference.
	Required Qty	Existing table field - calculated value	Displays the carton quantity required by the batch.
	On Hand Quantity	Existing table field	Displays available carton stock.
	Extra	Existing table action - updated function	Contains Issue Extra; disabled until the required carton issue is confirmed.
	Issue Extra	Existing button - updated controlled function	Opens additional carton issue. A positive quantity and reason are mandatory.
	Location Number	Existing table field	Displays the carton warehouse/location.
	Confirmed By	New audit table field	Displays Pending confirmation or the authenticated user and date/time.
	Done	Existing button -	Confirms the required carton issue and records attribution.

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3.5		Carton Issuing Module		
			updated controlled function	
	Completion count		New calculated indicator	Displays 0/1 or 1/1 completed.
	Back to Batch Queue		Existing button - common	Returns to the Carton Issuing queue.
Operations	The user opens a Reboxing batch that completed Leaflet Printing. The common product and controlled-document panels are displayed with the Carton to Issue table. The user verifies the carton reference, required quantity, on-hand quantity and location, then selects Done. The system records the required carton issue, changes the completion count to 1/1 and displays the authenticated user/date-time. Issue Extra becomes available only after the required issue is confirmed; it requires the extra quantity and a reason. The completed carton issue is retained in the BAR/audit history and the batch becomes eligible for Leaflet Folding once all other route-required work is complete.			
Use case	A Printer or Packaging user confirms the correct cartons and quantity have been issued for a reboxing batch.			
Output	Completed Carton to Issue table; required and extra carton quantities; location; reason; completion count; user/date-time attribution; BAR/audit update; and route-specific completion for Leaflet Folding eligibility.			
DS ID	3.5			
URS ID	4.2.6 - 4.2.8			
Business Rule	Only Reboxing batches that completed Leaflet Printing shall appear. Required carton issue shall be confirmed before Issue Extra is enabled. Extra quantity must be a positive whole number and requires a reason. Done shall be disabled after confirmation to prevent duplicate normal issue. Additional authorised issue shall use Issue Extra and remain auditable.			

3.6 Braille Printing Module

3.6		Braille Printing Module		
Priority		High		
Purpose		To print and complete the approved braille labels required for applicable relabelling batches after Leaflet Printing.		

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3.6	Braille Printing Module		
Role	Authorised Printer User		
Input	Field / Button / Table Column	Field type	Functional description
	Braille Printing List	Existing queue - updated eligibility	Lists Relabelling batches that completed Leaflet Printing and are flagged as requiring braille; uses section 3.2 common controls.
	Braille Labels to Print table	Existing table - updated function	Displays all required braille component lines and completion status.
	Braille Label	Existing table field	Rows include Braille Label and Braille Declaration Copy.
	Reference	Existing table field	Braille Label uses ECMA; Braille Declaration Copy uses PL No.
	Size	Existing table field	Displays pack size for the label and Master copy for the declaration.
	Quantity	Existing table field - calculated value	Displays braille quantity; declaration copy quantity is one.
	Location	Existing table field	Displays component location.
	In Hand Quantity	Existing table field	Displays available stock/on-hand quantity.
	Print	Existing button - common updated function	Opens the common Print window described in section 3.2.
	Mark as Done	New controlled button	Enabled when the required quantity for the line has been printed.
	Line Completion	New table status	Shows Pending, Mark as Done or Done.
	Completed By / Date	New audit table field	Displays the completing user and date/time.

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3.6		Braille Printing Module	
	Print Done	New controlled button	Enabled after every Braille Label and declaration row is Done.
	Back to Batch Queue	Existing button - common	Returns to the Braille Printing queue.
Operations	The user opens an eligible Relabelling batch for which braille is required. The common product and controlled-document panels are displayed. The Braille Labels to Print table includes the Braille Label and the Braille Declaration Copy. Each row follows the common process: verify reference, size, quantity, location and on-hand stock; perform the required test print; complete Actually Print; enter a reason for Extra Print; and mark the line Done. Each completion records user/date-time. Print Done is enabled only after all braille rows are complete, after which the batch becomes eligible for Leaflet Folding.		
Use case	A Printer user completes the required braille label lines for an eligible relabelling batch.		
Output	Completed braille label and declaration rows; test, actual and extra-print history; line-level attribution; final Braille Printing status; BAR/audit update; and route-specific completion for Leaflet Folding eligibility.		
DS ID	3.6		
URS ID	4.2.5 - 4.2.8		
Business Rule	The common controls in section 3.2 apply. Only eligible Relabelling batches with braille required and completed Leaflet Printing shall appear. All required quantities must be printed before line completion. The Braille Declaration Copy is a mandatory row where configured. Extra Print requires a reason. Print Done requires every displayed row to be Done.		

3.7 Leaflet Folding Module

3.7		Leaflet Folding Module
Priority	High	
Purpose	To complete and record leaflet folding after all printing and route-specific preparation required for the batch has finished.	
Role	Authorised Leaflet Folding User	

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3.7	Leaflet Folding Module		
Input	Field / Button / Table Column	Field type	Functional description
	B&S Batch Number	Existing search field	Searches the Leaflet Folding queue by B&S batch.
	MFG Lot No.	Existing search field	Searches the queue by manufacturing lot.
	Search	Existing button	Applies the Leaflet Folding search criteria.
	Leaflet Folding List table	Existing table - updated eligibility	Lists only batches that completed Leaflet Printing and all route-required Carton/Braille work.
	B&S Batch Number	Existing queue column	Displays the batch identifier.
	MFG Lot No.	Existing queue column	Displays the manufacturing lot.
	Product Name	Existing queue column	Displays product name.
	Strength	Existing queue column	Displays product strength.
	Pack Size	Existing queue column	Displays pack size.
	ECMA	Existing queue column	Displays approved reference.
	Expiry Date	Existing queue column	Displays batch expiry.
	Required Qty	Existing queue column	Displays the required leaflet/folding quantity.
	Status	New calculated queue column	Displays Active or the applicable controlled status.
	Select queue row	Existing table action - updated function	Opens the Leaflet Folding detail screen.

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3.7	Leaflet Folding Module		
	Product Information/Documents	Existing/common display - updated access	Uses the common product and controlled-document panel described in section 3.2.
	Leaflet Folding table	Existing table - updated function	Displays the leaflet identity, quantity and folding completion attribution.
	Leaflet Reference	Existing table field	Displays the approved ECMA/leaflet reference.
	Leaflet Size	Existing table field	Displays the approved pack-size leaflet format.
	Required Qty	Existing table field	Displays the quantity that must be folded.
	On Hand Quantity	Existing table field	Displays available leaflet quantity.
	Folded By	New audit table field	Displays Pending confirmation or the authenticated completing user.
	Date / Time	New audit table field	Displays Pending confirmation or the completion timestamp.
	Done	Existing button - updated controlled function	Completes Leaflet Folding, records folded quantity and attribution, and removes the batch from the active queue.
	Back to Batch Queue	Existing button	Returns to the Leaflet Folding list without completing the record.

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3.7	Leaflet Folding Module
Operations	The user searches the Leaflet Folding list by B&S Batch Number or MFG Lot No. The system shall list a batch only after Leaflet Printing and every route-required preparation activity are complete: Carton Issuing for Reboxing and Braille Printing for applicable Relabelling. The user selects the batch, reviews the common product/documents panel and verifies the Leaflet Reference, Leaflet Size, Required Qty and On Hand Quantity. Selecting Done records the folded quantity, authenticated user and date/time, updates the BAR/audit history, removes the batch from the active folding queue and hands it to pre-assembly as the next configured stage.
Use case	A Leaflet Folding user confirms the correct leaflet and batch, completes the required folding quantity and records completion.
Output	Filtered Leaflet Folding queue; verified batch/leaflet details; completed folding quantity; user/date-time attribution; updated BAR/audit history; removal from the active queue; and controlled handoff to pre-assembly.
DS ID	3.7
URS ID	4.3.1 - 4.3.6
Business Rule	Leaflet Printing completion alone is not sufficient. Reboxing requires completed Carton Issuing; Relabelling with braille required requires completed Braille Printing. Done requires valid batch identity, approved leaflet reference/format and required quantity. Completion is single-use in the normal workflow and shall be retained in batch history. The current wireframe queue logic shall be corrected to enforce these route-specific prerequisites.

4. Non-Functional Specification

This section provides complete URS-to-FS traceability and defines the non-functional controls for PLPI Batch Record Automation Phase 2, including audit, access, printing, testing, documentation and support.

URS ID & DS ID Ref	Function/Feature	Description/Specification
URS: 4.1.2 / DS: 3.1	B&S Batch Search and Selection	Search and select eligible products by batch, product, country, site, status and available criteria.
URS: 4.1.3 / DS: 3.1	Selected Batch Display	Display the product, pack, reference, expiry, quantity, order, location and route information needed to confirm the batch.

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URS ID & DS ID Ref	Function/Feature	Description/Specification
URS: 4.1.4 / DS: 3.1	Compatible Source Combination	Allow only compatible source records to be combined and retain traceability to each selected source.
URS: 4.1.5 / DS: 3.1	Controlled BAR Creation	Require confirmation of the selected B&S batch before creating one controlled electronic BAR.
URS: 4.1.6 / DS: 3.1	BAR Verification and Line Clearance	Provide batch, product, source-document and digital line-clearance checks.
URS: 4.1.7 / DS: 3.1	BAR Sign-off and Printing Handoff	Capture authenticated attribution, lock completed checks and route the batch to Label Printing.
URS: 4.2.1 / DS: 3.2	Printer Module Options	Provide Label Printing, Leaflet Printing, Carton Issuing and Braille Printing options.
URS: 4.2.2 / DS: 3.2	Printing Queue and Batch Display	Select a batch from the relevant queue and display route, component, quantity and location information.
URS: 4.2.3 / DS: 3.2	Controlled BAR and Artwork Access	Open the approved BAR, artwork and supporting documents for the selected activity.
URS: 4.2.4 / DS: 3.3	Route-Derived Label Lines	Display and require completion of all label lines applicable to the approved route.
URS: 4.2.5 / DS: 3.2 - 3.4, 3.6	Common Printing Process	Apply reference/quantity review, test or master checks, required and extra quantity controls and completion to Label, Leaflet and Braille Printing.
URS: 4.2.6 / DS: 3.4 - 3.6	Leaflet Completion and Route Branch	Route completed reboxing batches to Carton Issuing and applicable relabelling batches to Braille Printing.
URS: 4.2.7 / DS: 3.5, 3.6	Carton and Braille Route Controls	Confirm reboxing carton issue and complete braille labels only for eligible relabelling batches.

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URS ID & DS ID Ref	Function/Feature	Description/Specification
URS: 4.2.8 / DS: 3.2 - 3.7	Printing Completion and Folding Handoff	Record user/date-time, update status and move a route-complete batch to Leaflet Folding.
URS: 4.3.1 / DS: 3.7	Leaflet Folding Eligibility	List only batches that completed required printing and route-specific preparation.
URS: 4.3.2 / DS: 3.7	Folding Search and Batch Display	Search by B&S batch or manufacturing lot and display product, leaflet, quantity and status information.
URS: 4.3.3 / DS: 3.7	Folding Document Access	Provide controlled access to the BAR, approved leaflet and supporting documents.
URS: 4.3.4 / DS: 3.7	Batch and Leaflet Confirmation	Require agreement between the selected batch, leaflet and approved BAR.
URS: 4.3.5 / DS: 3.7	Folded Quantity and Completion Gate	Record folded quantity and prevent completion when identity, reference, format or quantity is missing or incorrect.
URS: 4.3.6 / DS: 3.7	Folding Attribution and History	Capture authenticated user, role and date-time and retain the completed folding record in batch history.

4.1 Audit Trail

The system shall retain audit records for B&S batch selection, source-record combination, BAR generation, viewing and printing, BAR and artwork access, continuation-page printing, B&S verification and line clearance, PDF opening, test/master review, required and extra print quantities, print category, reasons, component-line completion, Print Done, carton issue, braille completion, Leaflet Folding and every controlled status handoff. Audit entries shall include the batch, item/action, user, role where applicable, date/time and relevant quantity, reason or comments.

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4.2 Availability

The PLPI Phase 2 functions shall be available to authorised B&S Batch Add, Printer, Leaflet Folding, QA/RP, Operations and IT users during agreed operational hours.

4.3 Capacity Limits

The system shall support the expected operational volume of eligible B&S batches, generated BAR records, printing queue records, component lines, print runs, extra-quantity reasons, carton issues, braille records, folding records and retained audit history.

4.4 Performance

B&S Batch Add search, BAR generation and viewing, printing queue search, approved-document opening, print actions, quantity save, line completion, Print Done, carton issue and Leaflet Folding completion shall operate within agreed operational response times.

4.5 Recoverability

Saved BAR, printing and folding progress shall remain available for continuation. Completed BAR, line-clearance, printing, carton, braille, folding and workflow-handoff records shall recover without duplication or loss.

4.6 Security Requirements

Access shall be role based. B&S Batch Add users shall create and sign BAR records. Printer users shall access only the printing modules permitted by their role. Leaflet Folding users shall complete assigned folding records. QA/RP, Operations and authorised IT users shall have controlled review, exception, reporting or administration access. Completed records shall be read-only except through an authorised correction or reprint process.

4.7 Error Handling

The system shall validate eligibility, compatible selection, mandatory BAR/line-clearance checks, approved references, positive whole quantities, test/master controls, extra-print reasons and completion status. A failed validation shall block the action and retain the batch in its current controlled stage.

4.8 Usability

The workflow shall provide clear B&S Batch Add, printing and folding queues, consistent search and status presentation, readable controlled documents, line-level progress, visible completion attribution and clear route-based handoff messages without overlap or truncation.

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4.9 Accuracy & Validity

B&S identifiers, source selections, BAR data, approved references, quantities, reasons, checks, print runs, carton issues, braille and folding records shall remain linked through controlled identifiers and reflect the committed batch record.

4.10 User Access and Responsibilities

B&S Batch Add users select eligible records, generate the BAR, complete verification and line clearance and sign the batch to Label Printing. Printer users complete Label, Leaflet, Carton and Braille activities applicable to the route. Leaflet Folding users complete the folding record. QA/RP, Operations and IT users perform their authorised review, exception and support responsibilities.

4.11 Testing

Testing shall cover B&S eligibility and source compatibility; BAR creation, duplicate prevention, documents and line-clearance gates; printing menus, queues and common controls; test/master, actual and extra prints; quantities/reasons; Carton and Braille routing; line and Print Done gates; attribution; Leaflet Folding; audit history; errors, corrections, reprints and downstream handoff.

4.12 Controlled Documents and Training

Applicable SOPs, work instructions and training shall describe B&S Batch Add, BAR creation and verification, line clearance, the Printer menu and queues, Label/Leaflet/Braille common controls, test/master review, quantities and reasons, Carton Issuing, Leaflet Folding, controlled corrections, reprints and electronic sign-off.

4.13 Support and Administration

IT shall support PLPI access, configuration, printer and device connectivity, document and artwork links, workflow queues, audit retrieval, backup/recovery and incident resolution. Business, QA/RP and Operations shall own process decisions, approved references, master data, exception decisions and controlled use.